

Abaid-ur-Rehman

Machine Learning Engineer | AI Engineer | Python Developer

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SUMMARY

Machine Learning Engineer and Python Developer with hands-on experience in AI, Machine Learning, Computer Vision, and LLM-based applications. Skilled in Python, Flask, TensorFlow, Scikit-learn, OpenCV, and Retrieval-Augmented Generation (RAG). Passionate about building intelligent solutions and applying AI to solve real-world problems.

EDUCATION

Bachelor of Science in Artificial Intelligence, <i>NFC Institute of Engineering and Fertilizer Research, Faisalabad</i>	2023 – Cont.
Intermediate (Pre-Engineering), <i>KIPS College, Faisalabad</i>	2021 – 2023
Matriculation (Science), <i>Allied School, Faisalabad</i>	2019 – 2021

PROJECTS

Jarvis AI Desktop Assistant

Technologies: Python, Flask, Ollama , Speech Recognition

- Developed an AI-powered desktop assistant with voice command recognition and conversational AI capabilities.
- Implemented task automation including application control, web search, file management, and system utilities using local LLMs .

Smart Waste Classification System

Technologies: TensorFlow, Flask, OpenCV, Python

- Developed a deep learning application to classify waste into six categories using Convolutional Neural Networks (CNNs).
- Implemented image upload, real-time camera detection, and recycling recommendations through a responsive Flask web application.

Credit Card Fraud Detection System

Technologies: Python, Scikit-learn, XGBoost, Flask, Pandas

- Developed a machine learning system to detect fraudulent credit card transactions using multiple classification algorithms.
- Improved model performance through data preprocessing, feature engineering, and class imbalance handling techniques.

AI-Powered People Counting System

Technologies: YOLOv8, OpenCV, Flask, Python

- Developed a real-time people counting system using YOLOv8 object detection and multi-object tracking.
- Built a web dashboard to visualize occupancy statistics and crowd analytics in real time.

Apple Disease Detection

Technologies: TensorFlow, Keras, Flask, Python, NumPy, Pillow

- Built a CNN-based application to detect apple leaf diseases from images with high prediction accuracy.
- Designed a Flask dashboard displaying prediction results, confidence scores, and disease information.

AI-Image-Generator

Technologies: Python, Flask, JavaScript, HTML5, CSS3, Pillow, Hugging Face API, JSON

- Developed a full-stack AI image generation application integrating Hugging Face models for text-to-image generation.
- Implemented prompt enhancement, image editing (rotate, flip, brightness, contrast, saturation, sharpness), image history, favorites, search, sorting, and metadata management within a responsive user interface.

AI Typing Speed Tester

Technologies : Python , Flask , Supabase , PostgreSQL , JavaScript , Chart.js , HTML , CSS

- Developed a full-stack typing speed platform with secure authentication, real-time WPM tracking, analytics, leaderboards, achievements, daily & weekly challenges, user profiles, and XP-based gamification using Flask and Supabase.

SKILLS

Programming Languages

- Python
- SQL
- JavaScript
- HTML
- CSS

Soft Skills

- Problem Solving
- Analytical Thinking
- Team Collaboration
- Communication
- Continuous Learning

Machine Learning

- Machine Learning
- Deep Learning
- Supervised Learning
- Feature Engineering
- Data Preprocessing
- Model Training
- Model Evaluation
- Predictive Modeling

Database & Backend

- Supabase
- SQL
- REST APIs

Artificial Intelligence

- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- Natural Language Processing (NLP)
- Speech Processing
- Computer Vision
- Object Detection
- Image Classification
- Prompt Engineering
- AI Chatbot Development

Libraries & Frameworks

- Flask
- Streamlit
- Scikit-learn
- TensorFlow
- Keras
- Pandas
- NumPy
- OpenCV

Tools & Platform

- Git
- GitHub
- VS Code
- Jupyter Notebook
- Google Colab
- Kaggle

CERTIFICATES

Artificial Intelligence & Machine Learning with Python — NAVTTC

ACHIEVEMENTS

- Developed AI, Machine Learning, Computer Vision, and Full-Stack Python applications.
- Built real-time AI solutions including fraud detection, object detection, AI assistants, and image generation systems.
- Integrated modern AI technologies such as Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), YOLOv8, and Hugging Face APIs into practical applications.
- Continuously expanding expertise in Artificial Intelligence, Machine Learning, and Generative AI through hands-on projects.